Unit 3 Scale Exam

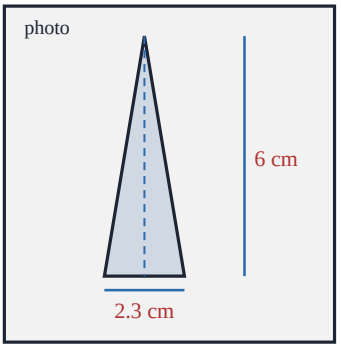
Math 9E Part 3 file · Lesson 3.4 scale and scale diagrams 6 questions · 20 marks Name: _____

The most important questions predicted from this worksheet file. Show all your work. Answers are on the last page.

Question 1

[3 marks]

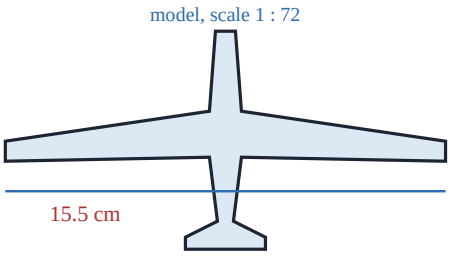
A photo shows a tower. On the photo the tower is 6 cm tall and its base is 2.3 cm wide. The real tower is 324 m tall. (a) Write the scale of the photo as a ratio 1 : n. (b) Find the real width of the base.



Question 2

[2 marks]

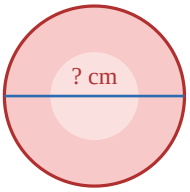
A collectable model plane has scale 1 : 72. The model's wingspan is 15.5 cm. Find the real wingspan in the most appropriate SI unit.



Question 3

[2 marks]

A red blood cell has diameter 5 μm. It is drawn at a scale of 8000 : 1. What diameter should the drawing have, in cm?



real cell: 5 μm

Question 4

[4 marks]

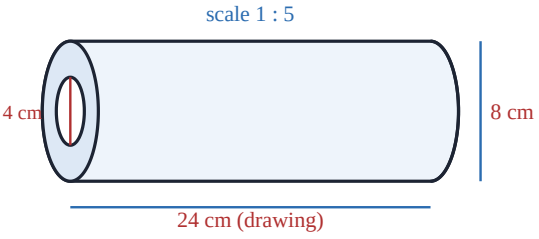
A map has scale 1:25 000. (a) Two towns are 18 cm apart on the map. How far apart are they in km? (b) A park covers 12 cm^2 on the map. Find its actual area in m^2 and in hectares.

(Careful: area does not use the same factor!)

Question 5

[4 marks]

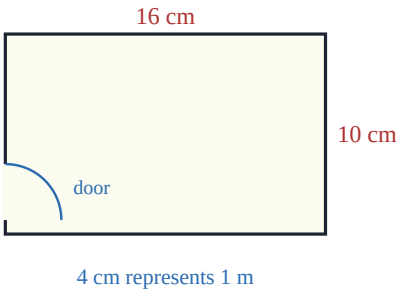
A scale diagram of a pipe has scale 1 : 5. On the diagram the pipe is 24 cm long, its outer diameter is 8 cm and the circular opening is 4 cm across. Find (a) the real length, (b) the real diameter of the opening and (c) the real volume of material in the pipe.



Question 6

[5 marks]

A bathroom is drawn on grid paper where 4 cm represents 1 m. On the plan it is 16 cm × 10 cm. The walls are 3 m tall and have one door of 0.9 m × 2.1 m that is not painted. Tiling costs \$225/m² and painting costs \$16/m². Find the total cost including 12% tax.



4 cm represents 1 m

ANSWER KEY

Q1 (a) 6 cm : 32 400 cm ⇒ **1 : 5400**. (b) 2.3 × 5400 = 12 420 cm = **124.2 m**.

Q2 15.5 × 72 = 1116 cm = **11.16 m**.

Q3 5 µm × 8000 = 40 000 µm = 40 mm = **4 cm**.

Q4 (a) 18 × 25 000 = 450 000 cm = **4.5 km**. (b) Area factor = $25\,000^2 = 6.25 \times 10^8$. 12 cm^2 × 6.25×10^8 = 7.5×10^9 cm^2 = **750 000 m^2 = 75 ha**.

Q5 (a) 24 × 5 = **120 cm**. (b) 4 × 5 = **20 cm** (outer diameter 40 cm). (c) $V = \pi(20^2 - 10^2)(120) = 36\,000\pi \approx$ **113 097 cm^3 ≈ 113.1 L**.

Q6 Room 4 m × 2.5 m. Floor = 10 m^2 × \$225 = \$2250. Walls: perimeter 13 m × 3 m = 39 m^2 − 1.89 m^2 (door) = 37.11 m^2 × \$16 = \$593.76. Subtotal \$2843.76; with tax ×1.12 = **\$3185.01**.

WHERE EACH QUESTION COMES FROM

Q1	Practice 3.4 #4 (Golden Gate photo scale)
Q2	Practice 3.4 #3 (toy plane 1:72)
Q3	Practice 3.4 #2 (red blood cell 8000:1)
Q4	Practice 3.4 #1 (map scale) plus area scale k^2
Q5	Practice 3.4 #7 ★★★ (pipe scale 1:5 and volume)
Q6	Practice 3.4 #8 ★★★★ (bathroom cost)